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PERFORMANCE SPECIFICATION

REFRIGERATION UNITS, ELECTRIC, 9000 BTU FOR MILITARY VAN CONTAINER, REFRIGERATED, INSULATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of insulated, electric refrigeration unit having a capacity of 9000 British thermal units (Btu) per hour and used with military van containers (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Government documents.

2.2.1 Standards. The following standards form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility
MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-1474	- Noise Limits for Military Materiel (Metric)

(Unless otherwise indicated, copies of the above standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standards Publication No. 250 - Enclosures for Electrical Equipment

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 No. 17th Street, Suite 1847, Rosslyn, VA 22209).

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, One Battery March Park, Quincy, MA 02269-9101).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

Standard 1496-2 - Thermal Containers

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

UNDERWRITERS LABORATORIES INC. (UL)

Standard 207 - Refrigerant Containing Components and Accessories, Nonelectrical
Standard 303 - Refrigerant and Air Conditioning Condensing and Compressor Units
Standard 353 - Limit Controls

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 **Operating requirements**. Refrigeration units shall satisfy the following minimum functional, operational and performance requirements.

3.2.1 4-hour operation. Refrigeration units shall operate properly with suction and discharge pressures safely controlled.

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3.2.2 72-hour operation. Refrigeration units shall operate properly and maintain proper conditions at thermostat settings of 0^{0}F and 35^{0}F.

3.2.3 Refrigeration capacity. Refrigeration units shall have minimum net capacities of 9,000 Btu/hr at a thermostat setting of 0^{0}F and 11,000 Btu/hr at a thermostat setting of 35^{0}F.

3.2.4 Heating capacity. Refrigeration units shall have a gross heating capacity of not less than 7,000 Btu/hr at a thermostat setting not less than 45^{0}F.

3.2.5 Defrost system. Refrigeration units shall have a defrost cycle terminated by temperature, automatically initiated by a time control adjustable between 6 and 24 hours and have the capacity to produce not less than 10 pounds of condensate within 30 minutes. The defrost system shall have provisions for manual defrost override.

3.2.6 Drainage system. Refrigeration units shall have a drainage system for the collection and discharge of condensate to the outside of the unit.

3.2.7 Moisture content. The moisture content of the refrigeration system shall be kept at a minimum to maintain proper operation of the refrigeration unit.

3.2.8 Refrigerant leakage. Refrigerant containing components shall be gastight.

3.2.9 Control Panel. The refrigeration units shall have a control panel for the operator having, but not limited to, the following items:

1. Start, run, stop control
2. Operation indicator
3. Refrigerator internal temperature control
4. Refrigeration unit running time indicator
5. High and low side pressure indicators
6. Refrigerator interior temperature indicator
7. Manual defrost control
8. Defrost operation indicator
9. Heating cycle operation indicator
10. Refrigeration cycle operation indicator
11. Power supply circuit breaker

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3.2.10 Ozone-depleting substances. The refrigerant shall be commercially available nonozone-depleting, and environmentally safe. Class I ozone-depleting substances, as determined by the Environmental Protection Agency (EPA), shall not be contained in, and shall be prohibited from use during the fabrication of the refrigeration unit (see 6.6).

3.2.11 Temperature control. The selected temperature of the refrigeration units shall be controlled automatically. The temperature control shall allow an incremental selection of normal refrigerator/freezer temperatures. When temperature settings are below 20^{0}F, the heating cycle shall not operate.

3.2.12 Power requirements. Unless otherwise specified (see 6.2), the refrigeration unit shall operate on 208/230 volt, 3-phase, 60-hertz (Hz) and 200 volt, 3-phase, 50-Hz power.

3.2.12.1 Power supply cable. The refrigeration unit shall be supplied with a power supply cable capable of operating properly with the power source and permanently attached to the outside of the unit.

3.2.12.2 Lighting power supply cable. The refrigeration unit shall be supplied with a minimum 46-foot, SO type cable for 110 volt container lighting permanently attached to the outside of the refrigeration unit.

3.2.13 Finish.. The refrigeration units, except where detrimental to proper operation, shall have a nontoxic chemical agent resistant green finish (see 6.7.1). The word "CARC", together with the month and year of finishing, shall be stencilled near the identification plate in block letters not less than 1 inch high using lusterless black chemical agent resistant paint (see 6.7.2). All paint shall be properly applied.

3.2.14 Safety. The refrigeration units shall minimize flammability, operational, and personnel hazards.

3.2.15 Airflow. The air exiting the evaporator shall be directed to create proper circulation inside the refrigerator it is used with.

3.3 **Support or ownership requirements**. Refrigeration units shall meet the following requirements.

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3.3.1 Transportability. Refrigeration units shall be able to be handled by a forklift truck and by lifting eyes or similar devices located on the top of the unit.

3.3.2 Design reliability. Refrigeration units shall be reliable and designed to operate without failure when subjected to the design reliability test.

3.3.3 Service and maintenance. All major components shall be accessible for adjustment, maintenance, servicing, and replacement without removing any other major component. All other components and assemblies shall be designed for ease of installation and replacement and shall be accessible for maintenance, repair, service and replacement with minimum disturbance to adjacent components. No special tools shall be needed for any maintenance, repair, service, installation or replacement action. Condenser and evaporator fins shall be protected from any damage that might occur during servicing or maintenance actions. Operator personnel shall be able to visually check the moisture content of the refrigeration system on all refrigeration units. All points requiring lubrication shall be equipped with standard fittings and shall be accessible from the outside of the unit. Each unit shall have the fittings or location tagged to indicate the temperature and grade of lubricant used and recommended frequency of lubrication.

3.3.4 Marking. Identification, instruction, control and gage plates shall be of corrosion-resistant material permanently attached to the outside of the refrigeration unit. Numbering and lettering shall be legible and permanently inscribed and shall be no less than 1/16 inch in height. All plates shall blend with the framed enclosure finish and shall have a nonreflective surface.

3.3 4.1 Identification plate. The identification plate shall contain the following information:

1. U.S.
2. Military model number and nomenclature
3. National Stock Number (NSN)
4. Serial number
5. Contract number
6. Manufacturer's name and address
7. Manufacturer's model number
8. Commercial and Government Entity (CAGE) number
9. Technical manual number, if applicable, (see 6.2 and 6.5)
10. Weight

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11. Refrigerant used

3.3.4.2 Instruction plates. Instruction plates shall contain the wiring and refrigeration cycle diagrams and instructions for start-up, operation, shutdown and preventive maintenance.

3.3.4.3 Control and gage nameplates. All controls and gages shall have identifying nameplates.

3.3.4.4 Electrical wiring. Electrical wiring shall be marked to allow identification with the wiring diagram.

3.3.5 Interchangeability. Component parts of the refrigeration unit which are replaceable shall be interchangeable with the same parts of other units furnished under the same contract.

3.3.6 Weight. The weight of the refrigeration units shall not be greater than 1000 pounds including the refrigerant and all necessary lubricants.

3.4 **Environmental requirements**. Refrigeration units shall operate under the following environmental conditions.

3.4.1 Ambient temperature. The refrigeration units shall start and operate in an ambient temperature of 125°F.

3.4.2 Rainfall. The refrigeration unit shall operate in rainfall.

3.4.3 Shock and impact. The refrigeration units shall be operable after being subjected to shock and impact.

3.4.4 Electromagnetic compatibility. When specified (see 6.2), the refrigeration units shall not exceed radiated emission (RE)102 limits for ground applications (Navy fixed and AF) over a range of 2 megahertz (MHz) to 400 MHz or the conducted emission (CE)102 limits. Short duration interference shall be exempt.

3.4.5 Storage conditions. The refrigeration units shall remain operable after storage in temperatures of minus 65°F to plus 155°F.

3.4.6 Noise level. The noise levels at personnel "operator's" and "servicing" positions shall not damage personnel hearing.

3.4.7 Corrosion resistance. The refrigeration units shall operate after exposure to a salt fog atmosphere.

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3.5 **Interface and interoperability requirements**. The refrigeration units shall interface with refrigerated containers (NSN 8115-01-084-5491)(see 6.1), and shall have interface dimensions as shown on figure 1 with the overall dimensions no greater than those shown on figure 1.

3.5.1 Government generators. The refrigeration units shall start and operate on a generator set, 10 kilowatt (KW), 60Hz, model MEP-003A, NSN 6115-00-465-1030 and the Tactical Quiet Generator set TQG 803A, NSN 6115-01-275-5061.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Conformance inspection (see 4.1.2)

4.1.1 First article inspection. When a first article inspection is required (see 3.1 and 6.2), a sample shall be examined for all the verifications listed in table I.

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TABLE I. Verification methods

Title	Requirement	Verification	Title	Requirement	Verification
OPERATING REQUIREMENTS	3.2	4.3.2	Transportability	3.3.1	4.3.3.1
4-hour operation	3.2.1	4.3.2.1	Design reliability	3.3.2	4.3.3.2
72-hour operation	3.2.2	4.3.2.2	Service and maintenance	3.3.3	4.3.3.3
Refrigeration capacity ($0^{o}F$)	3.2.3	4.3.2.3.2	Marking	3.3.4	4.3.3.4
Refrigeration capacity ($35^{o}F$)	3.2.3	4.3.2.3.3	Identification plate	3.3.4.1	4.3.3.4.1
Heating capacity	3.2.4	4.3.2.4	Instruction plates	3.3.4.2	4.3.3.4.2
Defrost system	3.2.5	4.3.2.5	Control and gage nameplates	3.3.4.3	4.3.3.4.3
Drainage system	3.2.6	4.3.2.6	Electrical wiring	3.3.4.4	4.3.3.4.4
Moisture content	3.2.7	4.3.2.1	Interchangeability	3.3.5	4.3.3.5
Refrigerant leakage	3.2.8	4.3.2.1	Weight	3.3.6	4.3.3.6
Control panel	3.2.9	4.3.2.7	**ENVIRONMENTAL REQUIREMENTS**	3.4	4.3.4
Ozone-depleting substances	3.2.10	4.3.2.8	Ambient temperature	3.4.1	4.3.4.1
Temperature control	3.2.11	4.3.2.9	Rainfall	3.4.2	4.3.4.2
Power requirements	3.2.12	4.3.2.10	Shock and impact	3.4.3	4.3.4.3 and 4.3.4.4
Power supply cable	3.2.12.1	4.3.2.10.1	Electromagnetic compatibility	3.4.4	4.3.4.5
Lighting power supply cable	3.2.12.2	4.3.2.10.2	Storage conditions	3.4.5	4.3.4.6
Finish	3.2.13	4.3.2.11	Noise level	3.4.6	4.3.4.7
Safety	3.2.14	4.3.2.12	Corrosion resistance	3.4.7	4.3.4.8
Airflow	3.2.15	4.3.2.1	**INTERFACE AND INTEROPERABILITY REQUIREMENTS**	3.5	4.3.5
SUPPORT OR OWNERSHIP REQUIREMENTS	3.3	4.3.3	Government generators	3.5.1	4.3.5.1

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4.1.2 Conformance inspection. Conformance inspection of the refrigeration units shall consist of all the verifications listed in table I, except the verification testing specified for the requirements of paragraphs 3.2.2 through 3.2.6, 3.3.1, 3.3.2, 3.4.1 through 3.4.7 and 3.5.1 shall not be done. Sampling for this inspection shall be as specified in the contract or order (see 6.2).

4.2 Order of verifications. Verification listed in table I can be performed in any order except verification tests shall be conducted in the order shown in table II.

TABLE II. Verification test sequence

Test sequence	Requirement Paragraph	Characteristic	Verification Paragraph
1	3.2.1	4-hour operation	4.3.2.1
2	3.2.2	72-hour operation	4.3.2.2
3	3.2.3	Refrigeration capacity	4.3.2.3
4	3.2.5	Defrost system	4.3.2.5
5	3.4.1	Ambient temperature	4.3.4.1
6	3.5.1	Generator operation	4.3.5.1
7	3.2.4	Heating capacity	4.3.2.4
8	3.4.3	Shock	4.3.4.3
9	3.4.3	Impact	4.3.4.4
10	3.3.1	Transportability	4.3.3.1
11	3.4.2	Rainfall	4.3.4.2
12	3.4.4	Electromagnetic compatibility	4.3.4.5
13	3.4.5	Storage conditions	4.3.4.6
14	3.4.6	Noise level	4.3.4.7
15	3.2.6	Drainage	4.3.2.6
16	3.4.7	Corrosion resistance	4.3.4.8
17	3.3.2	Design reliability	4.3.3.2.

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4.3 Verification methods. Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification and similarity to previously-approved or previously-qualified designs.

4.3.1 Verification alternatives. The contractor may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures to verify performance (see 6.4).

4.3.2 **Operating requirements verification**. All operating requirements shall be verified as follows.

4.3.2.1 4-hour operation test. The refrigeration unit evaporator shall be clamped to the evaporator opening of a calorimeter or other insulated structure. All refrigeration unit components shall be in place and the refrigeration unit doors shall be closed. The refrigeration unit shall be operated for 4 hours to determine that all mechanical and electrical components, including heater circuit operation, are functioning as intended and all controls are in calibration. The units shall have a heat load applied to the enclosure so that the unit operates more than 50 percent of the test time and cycles on the automatic controls three or more times per hour. Suction pressure at the compressor shall be verified to be within the established guidelines of the compressor manufacturer. Prior to completion of the 4-hour test, the air flow to the condenser shall be blocked off and the unit operated and automatic shutdown of the refrigeration unit verified when the compressor discharge limit is exceeded. The compressor discharge limit shall be that recommended by the compressor manufacturer for the refrigerant used. The air exiting the evaporator shall be checked to verify that the airflow direction is out and upward.

After a minimum operating time of 1 hour with the thermostat set at 0°F, the moisture indicator shall be observed to verify that the moisture content as shown by the component manufacturer's color chart data is not greater than 25 parts per million. A check for refrigerant leakage shall also be performed, with the unit inoperative, with a leak detector having a maximum sensitivity of 0.40 oz per year. Leakage shall be checked to verify that it is no greater than 0.5 oz per year. Verify that the compressor shaft seal leakage is not greater than 3.0 oz per year.

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4.3.2.2 72-hour operating test. A 72-hour test shall be conducted as follows:

a. The refrigeration unit shall be clamped to the evaporator opening of a calorimeter or other insulated structure.
b. A combined heat load of 5000 Btu/hr shall be impressed when the thermostat is set at 0°F and 6500 Btu/hr shall be impressed when the thermostat is at the 35°F setting.
c. The refrigeration unit shall be operated 36 hours with the thermostat set at 0°F and 36 hours with the thermostat set at 35°F. During this test, suction and discharge pressure, ambient air, condenser air, and evaporator in and out air temperatures shall be recorded on automatic recording instruments. Normal maintenance shall be performed during this test and verification shall be made that rated conditions are maintained during the 72-hour period in an ambient temperature of 110°F.

4.3.2.3 Refrigeration capacity tests.

4.3.2.3.1 Calorimeter calibration. The calorimeter calibration shall be run once prior to capacity tests to establish the coefficient of heat leakage to be used for capacity calculation. The coefficient of heat leakage for the calorimeter shall not exceed 25 Btu/hr/°F and the volume shall not be less than 150 cubic feet. A refrigeration unit shall be bolted to the evaporator opening in the calorimeter and the overall heat leakage coefficient shall be determined as follows:

a. An electric fan(s), of sufficient capacity to insure that no uncirculated air pocket will exist, shall be installed inside the calorimeter. The electrical energy of the fan motor shall be measured by the same method used for the heating elements in subparagraph c and this heat equivalent shall be added to other sources of heat added to the calorimeter.
b. Four thermocouples, shielded from the effects of radiant heat, shall be installed inside the calorimeter and positioned within 6 inches of the ceiling and floor at two opposite corners. The interior temperature shall be taken as the average of the four thermocouples.
c. Heating elements shall be installed approximately in the center of the calorimeter. Electrical heat input shall be measured by a watt-hour meter with a scale increment of 0.1 kilowatt-hour or less.

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d. Eight thermometers or thermocouples, shielded from the effects of the radiant heat, shall be installed approximately 18 inches from each corner of the calorimeter in the room where it is placed. The ambient temperature shall be taken as the average of these eight temperatures.

e. All openings in the calorimeter shall be sealed, and the calorimeter shall remain undisturbed throughout the test.

f. The ambient temperature shall then be maintained constant within $\pm$ 2^{0}F. Sufficient, steady, non-varying heat shall be added to the interior of the calorimeter to maintain a minimum of 60^{0}F and a maximum of 100^{0}F temperature difference between the interior and exterior of the calorimeter.

g. The test shall be considered concluded when eight consecutive half-hourly readings indicate that:

1. The ambient temperature has been maintained constant within $\pm$2^{0}F.
2. The temperature inside the calorimeter has been maintained constant within $\pm$ 1^{0}F.

h. The overall heat leakage coefficient U (Btu/hr/^{0}F) of the calorimeter is given by:

$$U = \frac{H_1}{(T_1 - T_2)}$$

H_1 = Sum of all electrical heat energies added to the calorimeter, Btu/hr.

T_1 = Average interior temperature of the calorimeter, (100^{0}F to 155^{0}F).

T_2 = Average ambient temperature, (T_1 - T_2) shall be in the range of 60^{0}F to 100^{0}F.

i. The test calorimeter shall not be used until a heat leakage coefficient no greater than 25 Btu/hr/^{0}F is obtained.

4.3.2.3.2 Capacity test (0^{0}F). The capacity of the refrigeration unit for low temperature shall be verified to be no less than 9000 Btu/hr using the method described below. Recorded data shall be required and shall include all specified temperature readings, electrical heat input, suction and discharge pressures, and complete calculations. All panels and doors of the refrigeration unit shall be in the closed position.

a. All accessories and appurtenances shall be in their proper place.

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b. Thermocouples, shielded from the effects of radiant heat, shall be installed in the air streams entering the evaporator and leaving the evaporator section.

c. Shielded thermocouples shall be located at the approximate center of four equal areas of the condenser face. The condenser inlet temperature will be taken as the average of these four temperatures.

d. Recording pressure gages shall be used for the compressor suction and discharge pressures.

e. The procedures described in 4.3.2.3.1 b, c, d, and e shall be followed for this test.

f. The refrigeration unit shall then be operated with air surrounding the condenser section and entering the condenser at a minimum of 110^0F, and sufficient, steady, non-varying heat shall be added to the interior of the calorimeter to maintain 0^0F temperature of air entering the evaporator, $\pm$ 1^0F. A watt-hour meter with scale increments of 0.1 kilowatt-hour or less shall be used to indicate the quantity of electrical energy added. Air velocity shall not exceed 100 feet per minute at any point 3 feet from the refrigeration unit.

g. The test shall be considered concluded when eight consecutive half-hourly readings indicate that:

1. The average calorimeter ambient temperature, T_3 has been maintained at steady state of $\pm$ 2^0F at a temperature of 100^0F to 110^0F.

2. Return air temperature, T_4, entering the evaporator has been maintained at 0^0 F $\pm$ 1^0F.

3. The net refrigerator capacity shall be considered as the sum of the electrical energy, in Btu/hr, added during this test plus the heat leakage gain in Btu/hr, as calculated from the calibration specified in 4.3.2.3.1.

The net refrigeration capacity is given by:

$Q = U\ (T_3 - T_4) + H_2$

Q = Net refrigeration capacity in Btu/hr

U = Overall heat leakage coefficient Btu/hr/^{0}F as determined by paragraph h in 4.3.2.3.1

H_2 = Sum of all heat energy added to the interior of the calorimeter in Btu/hr,from f above.

4.3.2.3.3 Capacity test (35^0F). The procedure specified in 4.3.2.3.2 shall be followed, with the following exceptions, to verify that the capacity of the refrigeration unit is not less than 11,000 Btu/hr.

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a. In 4.3.2.3.2 f, sufficient heat shall be added to maintain $35^0F \pm 1^0F$ temperature of air entering the evaporator in place of 0^0F temperature specified.

b. In 4.3.2.3.2 g 2, temperature of the air entering the evaporator shall be maintained at $35^0F \pm 1^0F$ in place of 0^0F temperature specified.

4.3.2.4 Heating capacity test. The following procedures shall be taken with the refrigeration unit in the calorimeter:

a. The internal temperature of the calorimeter shall be $30^0F \pm 5^0F$.
b. With fans running, the ambient temperature surrounding the condenser section shall be minus 40^0F.
c. A wattmeter shall be attached to read the wattage of the heater.
d. The thermostat shall be set at no less than 45^0F.
e. The unit shall be turned on and the wattage reading taken when the calorimeter reaches 40^0F.

The wattmeter shall read a minimum of 2050 watts.

4.3.2.5 Defrost test. With the refrigeration unit operating in a maximum ambient temperature of 20°F and the insulated structure maintained at $0^0F \pm 10°F$, the evaporator shall be frosted by spraying water on the coil surface or evaporating water in the box. After the coil is frosted, the refrigeration unit shall be set on the defrost cycle and the condensate collected and weighed. Verification shall be made that the unit operates through a complete defrost cycle, that no less than 10 pounds of condensate is produced in 30 minutes, that no overloading of the compressor motor occurs, that there is no damage to the compressor because of lack of lubrication or refrigerant "slugging" and that no condensate gets into the food storage area when the refrigerant cycle begins again. Verify that there is no ice on the coil at the end of the defrost cycle and that there are provisions for manual defrost override.

4.3.2.6 Drainage test. Water shall be added to the drainage system at the rate of 1 gallon every 4 to 5 minutes and the refrigeration unit shall be tilted 15 degrees fore-and-aft. While in both the fore-and-aft directions, observations shall be made to verify that condensate is collected and drained outside the condenser without spillage.

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4.3.2.7 Control panel inspection. During the 4-hour operation test, visually verify that all controls and indicators are functioning as intended.

4.3.2.8 Ozone-depleting chemicals inspection. Visually check publication SD-14 (see 6.6) to verify that the refrigerant used in the refrigeration unit is not listed.

4.3.2.9 Temperature control inspection. Visually inspect the temperature control to verify that temperatures from minus 10°F to plus 50°F can be selected in increments not less than 1°F and not greater than 2°F. Verify lockout of the heating cycle by operating the refrigeration unit in the calorimeter with the temperature control set at 18°F in an ambient temperature of not less than 50°F. When the temperature inside the calorimeter reaches 18°F, lower the ambient temperature to bring the calorimeter temperature below 18°F and verify that the heating cycle does not operate.

4.3.2.10 Power requirements demonstration. During the 72-hour operation test on the refrigeration units, observe that the refrigeration unit operates on a 208/230 volt, 3-phase, 60-Hz power source for one 36-hour period of operation and on 200 volt, 3-phase, 50-Hz power for the other 36-hour period.

4.3.2.10.1 Power supply cable certification. The power supply cable shall be certified as conforming to the requirements of ISO 1496-2, including length and storage requirements, with male and female connectors also certified as conforming to Annex M of ISO 1496-2. Visual verify that the cable is permanently attached to the outside of the refrigeration unit.

4.3.2.10.2 Lighting power supply cable examination. Verify that a minimum 46-foot, S0 type cable for 110 volt operation is permanently attached to the outside of the refrigeration unit.

4.3.2.11 Finish certification and examination. The contractor shall certify that the refrigeration unit's finish is nontoxic and chemical agent resistant, meeting Government requirements. Painted surfaces shall be visually examined to verify that there are no streaks, crops, pinholes, checks or cracks.

4.3.2.12 Safety certification and inspection. Safety requirements shall be verified a follows.

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4.3.2.12.1 Wiring certification. Wiring practices used in the refrigeration units shall be certified as conforming to NFPA Standard No. 70 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.12.2. Electrical enclosure certification. Electrical enclosures (junction boxes, control boxes, etc.) shall be certified as conforming to the watertightness requirements of NEMA Standards Publication No. 250 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.12.3 Refrigerant containing components certification. Refrigerant containing components used in the refrigeration units shall be certified as conforming to UL Standard 207; condensing units and compressors used shall be certified as conforming to the requirements of UL Standard 303 and limit controls used shall be certified as conforming to UL standard 353. Verification of certification shall be done by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.12.4 Personnel safety inspection. Visually verify that a plate indicating "Danger, Rotating Fan" has been permanently attached on both sides of the condenser.

4.3.3 **Support or ownership requirements verification**. All ownership and support requirements shall be verified as follows.

4.3.3.1 Transportability test. The refrigeration unit, loaded to 1-1/2 times the weight of the unit, shall be lifted, transported, and placed in position by a forklift truck without securing the unit in any way to the truck. During the transport, the truck shall be driven over a 4-inch by 4-inch board so either the two left or two right tires go over the board. After the forklift transport, the refrigeration unit shall be raised by its lifting eyes or similar devices and held suspended for 5 minutes. Upon completion, visually examine the unit to verify that there is no damage or distortion.

4.3.3.2 Design reliability test. A 2500-hour design reliability test shall be run as follows:

a. A maximum of two units shall be used for this test. When two units are used, each unit shall be tested for a minimum of 1000 operating hours and their combined testing time must equal 2500

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operating hours. Operating hours for the units shall be the hours the compressor is operating.

b. The refrigeration unit shall be clamped to the evaporator opening of the calorimeter (see 4.3.2.3.1) with a minimum volume of 150 cubic feet.

c. A heat load shall be imposed on the unit so the compressor will cycle with running time of approximately 75 percent when the condenser air entering temperature is 110°F.

d. The air entering the condenser and surrounding the condensing section shall be maintained at a minimum of 110°F for 90 percent of the operating hours and at a minimum of 125°F for 10 percent of the operating hours.

e. The defrost control shall be set for two defrosts in a 24-hour period of operating time.

f. The refrigeration shall be operated with the thermostat set at 0°F for 36 hours and at 40°F for the next 36 hours. This cycle hall be repeated until all operating hours have been completed. During the test, suction and discharge pressures, air temperature entering the condenser, refrigerated enclosure temperature, and compressor operating hours shall be recorded every hour on automated recording instruments.

g. Failure of the unit to maintain the temperature of the closure, failure to start and operate, failure to defrost, or any mechanical or control failure that requires more than 30 minutes to repair, shall be defined as a unit failure. A malfunction that can be repaired in less than 30 minutes shall be classified as an incident.

h. When failure to complete test hours occurs, the failure shall be analyzed, correction made, and the test started from time zero. All failures, analyses, and corrective actions shall be recorded.

i. The test shall be successfully completed when the unit has completed all operating hours without failure and with no more 5 incidents.

4.3.3.3 Service and maintenance simulation. A simulated demonstration shall be accomplished verifying that all major refrigeration unit components are accessible for adjustment, maintenance, servicing and replacement without the removal of any other major component or need for special tools. The simulation shall also illustrate that condenser and evaporator fins have guards or other protection and all other components and assemblies are accessible for maintenance, repair, service or replacement and

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require no special tools for those actions. Verification shall be made that the moisture content of the refrigeration system can be visually determined. Verify that lubrication fittings have tags identifying temperature and grade of lubricant and recommended frequency of lubrication.

4.3.3.3.1 Alternative service and maintenance analysis. When a technical manual is furnished as part of the contract, it can be analyzed for proper service and maintenance procedures in lieu of the simulation specified in 4.3.3.3.

4.3.3.4 Marking inspection. Visually verify that all identification, instruction, control and gage plates are permanently attached to the outside of the refrigeration unit, lettering is permanently inscribed, legible and not less than 1/16 inch high, and plates have a nonreflective surface.

4.3.3.4.1 Identification plate inspection. Visually verify that all information is complete and accurate.

4.3.3.4.2 Instruction plates inspection. Visually verify that all diagrams and instructions are complete and accurate.

4.3.3.4.3 Control and gage nameplate inspection. Visually verify that all controls and gages have nameplates.

4.3.3.4.4 Electrical wiring inspection. Visually verify that all wiring is marked to allow identification with the wiring diagram on the instruction plates.

4.3.3.5 Interchangeability certification. The contractor shall certify that all component parts of the refrigeration unit which are replaceable are interchangeable with the same parts of other units of the same type furnished under the same contract.

4.3.3.6 Weight examination. Verify that the weight of the refrigeration unit is 1000 pounds or less by weighing the entire unit with all necessary oils, lubricants and refrigerant in place.

4.3.4 **Environmental requirements verification**. All environmental requirements shall be verified as follows.

4.3.4.1 Ambient temperature test. The refrigeration unit shall be heat soaked for 6 hours at an ambient temperature of 125^{0}F minimum. With air entering and surrounding the condenser and the

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condenser section maintained at 125°F minimum, the unit shall be started and operated for 1 hour with its control set at 40°F, then shut off. Air velocity around the condenser section shall not exceed 100 feet per minute. After 2 minutes, the unit shall be restarted and operated for 1 hour with its control set at 40°F, then shut off. This procedure shall be followed for a total of 4 hours with the unit continuing to start and operate.

4.3.4.2 Rainfall test. The refrigeration unit shall be mounted in a simulated container and fresh water sprayed on the unit at no less than 3 inches per hour at an average downward angle of 45 degrees to simulate rainfall on the exposed surfaces of the unit. The front, top, and two sides of the unit shall be individually sprayed for 30 minutes while not operating with the spray applied perpendicular to the surfaces. The refrigeration unit shall then be started and operated for 30 minutes while the spray continues. A visual verification shall be done to insure no leakage occurred in the evaporator section, electrical control or junction boxes or the food storage area of the simulated container.

4.3.4.3 Shock test. The refrigeration unit shall be mounted on a drop-type frame, constructed so that the refrigeration unit can be mounted as it is installed on a refrigerated container. The drop frame shall be constructed of steel with the refrigeration unit supported by vertical members that transmit shock directly from the floor to the refrigeration unit. The refrigeration unit shall be tested as follows:

a. The drop frame shall be lifted vertically and free dropped ten times onto a concrete floor from a height of 6 inches. Upon completion of this test, verify that there has been no:

(1) Permanent distortion of 1/32 of an inch on the unit frame measured from center punch points located on both sides of the unit frame, one in the upper corner and one on the diagonally opposite lower corner.

(2) Sag in the evaporator section greater than 1/32 of an inch measured by scribing lines on the evaporator section parallel to the drop frame so that movement relative to the unit frame can be measured.

(3) Visible changes in the alignment of the condenser or evaporator fan and shroud, as applicable, greater than 1/8 of an inch.

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(4) Refrigerant leaks, observable damage, movement of misalignment of any components of the unit.

(5) Cuts or frayed electrical wire.

b. After successfully passing a, the unit, still mounted on the drop frame, shall be started and operated for 30 minutes and the unit tested again to verify that there are no leaks, malfunctioning components or controls, abnormal vibration, or other irregular operation.

4.3.4.4 Impact test. The refrigeration unit shall be mounted in the test frame used for the shock test. The unit shall be started and left running during this test. The test frame shall be raised and dropped so that the refrigeration unit receives an impact of 5G magnitude. This impact shall be repeated for a total of 200,000 times. Every 4 hours, tightening of loose nuts and bolts will be allowed and all corrections shall be recorded. After completion of the test, the refrigeration unit shall be visually examined to verify that there are no defects such as deformation, loose components, strain cracks in the unit frame, ruptured lines, leaks, and cut or frayed electrical wire. After passing the visual examination, the unit shall be started and, while operating, shall be examined to verify that there is no mechanical malfunction or leakage.

4.3.4.5 Electromagnetic compatibility tests. When electromagnetic compatibility is required (see 6.2), the refrigeration unit shall be tested in accordance with MIL-STD-462. The units shall meet the RE102 and CE102 limits of MIL-STD-461.

4.3.4.6 Storage test. The refrigeration unit shall be subjected to a temperature of 155°F for 4 hours, then the room temperature shall be brought to 70°F ± 5°F and the unit operated for 1 hour. The refrigeration unit shall then be subjected to a temperature of minus 65°F for 4 hours, then the room temperature shall be increased to 70°F ± 5°F and the unit operated for 1 hour. Verify that there has been no permanent set or loss of resiliency of packing or gaskets, congealing of lubricants, discoloration, cracking, bulging, checking or crazing.

4.3.4.7 Noise limit test. The refrigeration unit shall be tested for noise level as specified in MIL-STD-1474. The noise limit category shall be "D". The "operator's" position shall be 10 feet in front of the unit and 5 feet above the ground level. The

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"servicing" position shall be 2 feet in front of the control panel and 4 feet above the ground level. The noise level shall be less than 85 decibels.

4.3.4.8 Corrosion resistance test. The refrigeration unit shall be tested for its resistance to the effects of a salt fog atmosphere having a concentration of 5 percent ± 1 percent. The unit, in its operational configuration, shall be tested in accordance with MIL-STD-810, Method 509.3, with the refrigeration unit subjected to 24 hours of salt fog exposure followed by 24 hours of drying time in an ambient temperature of $70^{0}F \pm 5^{0}F$, followed by a second 24-hour exposure and 24-hour drying time. The unit shall then be operated and visually checked over all surfaces to verify that there is no corrosion.

4.3.5 **Interface and interoperability requirements verification.** The refrigeration units shall be measured to determine that interface dimensions are as shown on figure 1 and overall dimensions are no greater than indicated on figure 1.

4.3.5.1 Generator test. A government furnished generator set shall be provided for this test (see 3.5.1 and 6.2). The ambient temperature test specified in 4.3.4.1 shall be repeated with the refrigeration unit continuing to start and operate when operating on the government furnished generator set.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

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6.1 Intended use. The refrigeration units covered by this specification are intended for worldwide use by the Military, both in the field and at sea, under adverse climatic conditions without added protection from the elements. In use, the evaporator section of the refrigeration unit extends into the opening provided in the refrigerated container (NSN-8115-01-084-5491).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DoDISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3)
c. When a first article is required (see 3.1, 4.1.1 and 6.3)
d. When power requirements are other than specified (see 3.2.12).
e. Technical manual number, when required (see 3.3.4.1).
f. When electromagnetic compatibility is required (see 3.4.4 and 4.3.4.5)
g. Sampling for conformance inspection (see 4.1.2)
h. Government generator set to be furnished (see 4.3.5.1).
i. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously-acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Verification alternatives. Contracting documents should provide guidance to offerors regarding the submission of alternatives to specified verification methods (see 4.3.1).

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DoD 5010.12-L,

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Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under a separate contract line item in the contract.

6.6 Ozone-depleting substances. A listing of ozone-depleting substances prohibited from use can be found in publication SD-14 Listing of Toxic Chemicals, Hazardous Substances, and Ozone-Depleting Chemicals. It is available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.7 Finish.

6.7.1 Paint, green. Treatment and painting in accordance with MIL-C-53072 or types F or G of MIL-T-704 have been used to satisfy this requirement (see 3.2.13).

6.7.2 Paint, black. A paint in accordance with MIL-C-53039 or types III or IV of MIL-C-46168 have been used to satisfy this requirement (see 3.2.13).

6.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

6.9 Subject term (key word) listing.

Container, ISO
Container, thermal
Food preservation, field
Food storage, field
Refrigeration, field

6.10 Supersession data. MIL-R-0043891B (GL), which provides a detailed configuration of the refrigeration unit, is superseded by this performance specification. MIL-R-0043891B and associated drawings are available to the contractor on an information basis.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:

Army - GL
Navy - YD1
Air Force - 99

Review activities:

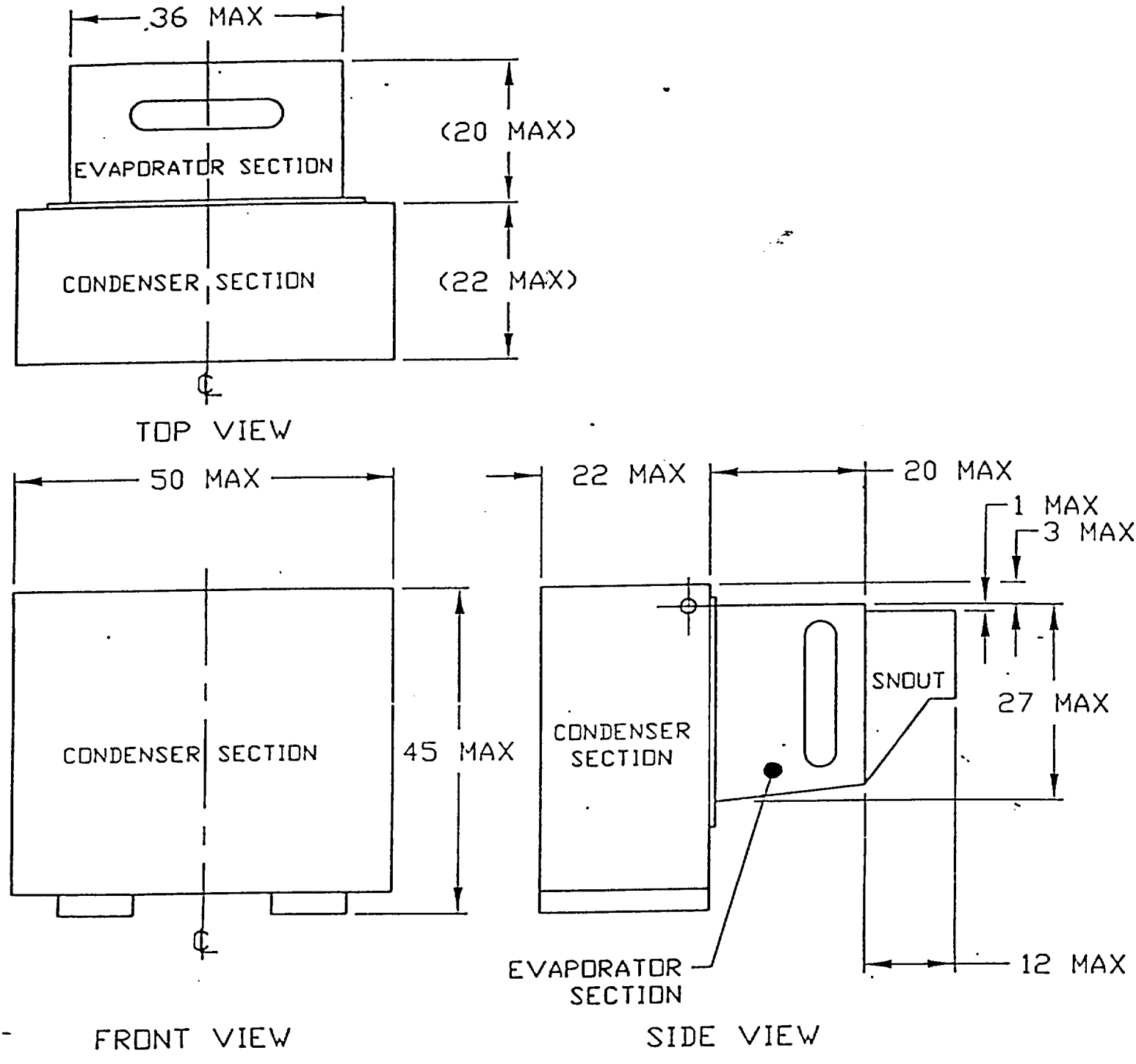
Army - AV, MD1, SM
Navy - MC
Air Force - 82
DLA - IS

Preparing activity:

Army - GL

(Project 4110-0501)

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NOTE:

1. UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES.

FIGURE 1.
DIMENSIONAL REQUIREMENTS FOR
9000 BTU REFRIGERATION UNIT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-43891C	2. DOCUMENT DATE (YYMMDD) 97-08-05

3. DOCUMENT TITLE Refrigeration Units, Electric, 9000 BTU for Military Van Container, Refrigerated, Insulated

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)
a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial 508-233-5175	(2) AUTOVON 256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center Attn: SSCNC-WEF Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: *Defense Quality and Standardization Office* 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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